Kangxian Xie

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Education

University at Buffalo, The State University of New York **Aug. 2024 - Present**
Doctor of Philosophy in Computer Science - GPA:3.95/4.0 *Buffalo, New York*

Boston University **Sep. 2021 – Jan. 2023**
Master of Science in Computer Science - GPA: 3.7/4.0 *Boston, Massachusetts*

University of California, San Diego **Sep. 2015 – Dec. 2019**
Bacholar of Science in Math and Computer Science *La Jolla, California*

Professional Experience

Research Assistant at Dep. of Comp. Sci. and Engineering | *University at Buffalo* **Aug. 2024 - Present**

- Currently focusing on Developing and testing AI models to analyze surgical videos. Building tools to identify instruments, describe actions, and track changes over time. Evaluating how well existing models perform on surgical footage.
- Developed an end-to-end vision-language framework that integrates 3D anatomical representations with natural-language instructions to iteratively refine 3D medical segmentations. Achieved significant improvements in segmentation accuracy, demonstrating the feasibility of precise, language-guided 3D anatomical correction. *MICCAI*, 2026 (Paper, Code).
- Conduct research on 3D medical image analysis and shape modeling, with focus on anatomically consistent reconstruction via 3D shape deformation. Proposed to combine deep learning-based template and learned deformation feature field for on 3D Pulmonary (Lung) Segment reconstruction. The proposed method achieved 4% enhancement in normalized surface dice over the previous SOTA.*Medical Image Analysis, 2026* (Paper).

Machine Learning Researcher | *Harvard University & EPFL & Boston College* **Nov. 2022 - Aug. 2024**

- Developed a novel deep learning framework for labeling complex pulmonary tree structures, achieving a 5% improvement in labeling accuracy and a 10% enhancement in computational efficiency compared to prior methods. The approach incorporated point&graph 3D representations and Implicit-Nerual-Representation modeling to enhance spatial understanding of the anatomy. *Medical Image Analysis*, 2025 (Paper, Code).
- Proposed a task-oriented augmentation framework improving biomedical segmentation models' performance by 3.4% for cell images and 1.3% for lung images. Introduced a novel scale-style selection mechanism leveraging consistency metrics for real-world deployment adaptability, optimizing both segmentation accuracy and domain generalization. *IEEE International Symposium on Biomedical Imaging (ISBI)*, 2024 (Paper, Code).

AI Engineer - Automatic Vision Inspection Project | *Nextas Technology Co.* **Oct. 2020 - Aug. 2021**

- Evaluated, designed, and debugged deep learning-based AI machine vision software for defect detection of electronic components, including camera modules in smartphones.
- Created and processed extensive defect image datasets, trained and fine-tuned deep learning models deployed on all sold inspection devices to ensure high accuracy and robustness.
- Contributed to achieving industry-leading inspection metrics: detection efficiency of 3000 pcs/hr (vs. industry standard of 1800 pcs/h), false negative rate $\leq$ 0.05% (vs. 0.2%), and false positive rate 2% (vs. 5%).

Projects

MRI-based Alzheimer's Disease Indicator Prediction | BILab@Boston Univ. **May. 2022 – Aug. 2022**

- Predict Alzheimer's disease bio-marker values using deep learning transformer-based models on 2D Brain MRI.
- Utilize an attention-rollout technique to identify brain sub-regions most responsible for prediction.

Computer Vision Challenge: Visual Odometer | Boston Univ. CSE 574 **Mar. 2022 – Apr. 2022**

- Achieve Top 3 accuracy in class of 70+ graduate students by applying iterative method (RANSAC) for feature matching and essential matrix refinement.

Reinforcement Learning for Optimal Pathfinding | Boston Univ. CSE 542 **Nov. 2021 – Dec. 2021**

- Implemented Q-learning, a reinforcement learning algorithm, to train an autonomous character to navigate a map while avoiding undesired terrains and reaching a goal location efficiently. Designed and optimized the reward function to balance exploration and exploitation.
- Conducted extensive hyper-parameter tuning (learning rate, discount factor) to achieve optimal convergence and robust performance across different map configurations. Visualized training progress and policy effectiveness through simulation environments, demonstrating the algorithm's adaptability to varying obstacle distributions.

Skills

Technical Skills: pytorch, calculus, abstract algebra, data structure and algorithms, Latex, Microsoft Office.
Programming Languages: Python, R, Java, C++.